

**MINISTRY OF EDUCATION AND TRAINING  
FPT UNIVERSITY**

**A TWO-TIER COGNITIVE ARCHITECTURE FOR  
AUTOMATED THREAT DETECTION AND RESPONSE  
USING AGENTIC AI**

by

Nguyen Duc Binh

A thesis submitted in conformity with the requirements  
for the degree of Master of Software Engineering

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Supervisor:  
Dr. Bui Van Hieu  
Dr. Dang Van Hieu

## Abstract

In the modern cybersecurity landscape, Security Operations Centers (SOCs) face mounting pressure to automate the triage and response of an ever-increasing volume of security alerts; yet every available solution entails a significant trade-off. Conventional automation tools operate through rigid, pre-scripted playbooks that can neither reason about ambiguous situations nor justify their decisions, whereas feeding raw logs directly to a Large Language Model (LLM) proves too slow for real-time operation, exposes the system to manipulation through the log content itself, and violates data-sovereignty requirements when deployed in the cloud. To overcome these limitations, this thesis proposes SENTINEL, a two-tier cognitive architecture for automated threat detection and response. The first tier is a fast, deterministic filter that instantly discards the majority of benign traffic and forwards only the genuinely suspicious minority to the second tier: a locally hosted artificial-intelligence agent that reasons over authoritative security knowledge sources (MITRE ATT&CK techniques and NIST incident-response guidance) and operates behind multiple protective guardrails to withstand deception by malicious inputs.

Evaluated on standard intrusion-detection datasets, SENTINEL demonstrates that AI-based reasoning can remain practically viable: by reserving this costly computation for the small fraction of alerts that genuinely warrant it, the system reduces average response latency by more than 80% relative to an LLM-only approach. It further withstands content-based manipulation attacks, produces transparent and verifiable justifications for each decision, and runs entirely on local, consumer-grade hardware without transmitting any data off-site. Taken together, these findings affirm that pairing a high-speed deterministic filter with a secure, transparent, and locally deployed AI reasoning tier constitutes a practical and trustworthy approach to automating modern network defence.

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## List of Abbreviations

| Abbreviation | Full Description                           |
|--------------|--------------------------------------------|
| APT          | Advanced Persistent Threat                 |
| CTI          | Cyber Threat Intelligence                  |
| DAPT         | Dynamic/Delayed Advanced Persistent Threat |
| FAISS        | Facebook AI Similarity Search              |
| GGUF         | GPT-Generated Unified Format               |
| HITL         | Human-in-the-Loop                          |
| LLM          | Large Language Model                       |
| RAG          | Retrieval-Augmented Generation             |
| RRF          | Reciprocal Rank Fusion                     |
| SIEM         | Security Information and Event Management  |
| SOC          | Security Operations Center                 |

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# Chapter 1

## Introduction

### 1.1 Background and Problem Statement

In the modern cybersecurity landscape, Security Operations Centers (SOCs) are grappling with a fundamental issue: **Alert Fatigue**. First-line defense systems (IDS/IPS, Firewalls) continuously generate hundreds of thousands of alerts daily. Due to their reliance on static rules or signatures, these systems lack contextual reasoning capabilities, resulting in an exceptionally high rate of false positives.

Large Language Models (LLMs) offer immense potential for contextual analysis, but deploying them directly into a network pipeline faces three critical barriers:

1. **Prohibitive Latency:** Using LLMs to analyze every high-speed network log in real-time is computationally infeasible.
2. **Manipulation Risks (Prompt Injection):** Attackers can inject malicious instructions directly into the payload (e.g., HTTP User-Agent) to deceive the LLM and bypass detection.
3. **Data Sovereignty:** Organizations cannot transmit sensitive network logs to third-party APIs (such as cloud services) due to strict Zero-Trust security requirements.

### 1.2 Solution: The SENTINEL Two-Tier Architecture

To decisively address these challenges, this thesis proposes **SENTINEL** – a two-tier cognitive architecture combining a high-speed filter with a locally deployed Agentic AI:

- **Tier-1 (Statistical Filter & ML Gateway):** Processes massive traffic volumes at wire speed. The ML Gateway (LightGBM) automatically filters out benign data and blocks overt attacks. Only a small minority of “ambiguous” cases are escalated to

Tier-2. Additionally, a **Semantic Cache (Tier 1.75)** instantly blocks repeated attacks (e.g., DDoS, Brute-force) within 1 millisecond without invoking the LLM.

- **Tier-2 (Cognitive LLM Agent):** Employs a locally hosted, quantized LLM (Gemma-9B). It leverages a RAG (Retrieval-Augmented Generation) knowledge base, grounded in the MITRE ATT&CK framework, to evaluate evidence and issue verdicts. Crucially, the LLM does not autonomously dictate actions; its outputs are constrained by a **Unified Decision Policy**, ensuring transparency and predictability.

This approach demonstrates that integrating deep AI reasoning into network defense is highly practical, cost-effective, resilient against evidence manipulation, and strictly adheres to data privacy constraints.

### 1.3 Thesis Structure

Chapter 2 summarizes the core theoretical foundations (Agentic AI, RAG). Chapter 3 details the architecture of the SENTINEL system. Chapter 4 evaluates the experimental results regarding accuracy and system performance. Chapter 5 concludes the thesis and outlines future development directions.

## Chapter 2

# Theoretical Background

This chapter outlines the core technologies forming the SENTINEL architecture: Agentic AI, Retrieval-Augmented Generation (RAG), and Zero Trust principles.

### 2.1 Agentic AI and LLM Quantization

Unlike traditional Runbook Automation tools that follow rigid, pre-defined scripts, Agentic AI systems possess the autonomy to analyze contexts, evaluate evidence, and make flexible decisions based on live data streams.

However, running Large Language Models (LLMs) locally requires massive hardware resources. To make Agentic AI practically viable for enterprises, SENTINEL applies **Quantization** techniques. By reducing the precision of model weights from FP16 (16-bit) to the Q6\_K standard (approx. 6.6-bit) via the GGUF format, the VRAM requirement for the Gemma-9B model shrinks from  $\approx 18$  GB to merely  $\approx 7 - 8$  GB. This allows smooth inference on consumer-grade hardware with negligible cognitive degradation.

### 2.2 Hybrid Retrieval-Augmented Generation (RAG)

LLMs are inherently prone to "hallucinations" – generating plausible but fabricated information. To counteract this, RAG forces the LLM to ground its reasoning on an authoritative knowledge base (e.g., MITRE ATT&CK) before responding. SENTINEL utilizes a **Hybrid Search** methodology that combines:

- **Vector Search (Semantic Search - FAISS):** Captures semantic similarities in attack descriptions.
- **Keyword Search (BM25):** Precisely matches technical identifiers and IP addresses.

The results from both methods are merged using Reciprocal Rank Fusion (RRF), yielding

optimal retrieval accuracy.

## 2.3 Zero Trust Security and LLM Vulnerabilities

The Zero Trust principle strictly prohibits transmitting sensitive network telemetry outside the organizational perimeter. Consequently, the entire SENTINEL architecture must operate in an Air-gapped (fully local) environment.

Furthermore, reading network logs with LLMs introduces a novel vulnerability: **Log-background Prompt Injection**. Attackers can deliberately inject deceptive control commands into the Payload (e.g., HTTP User-Agent). Without strict "Guardrails" to encapsulate the Payload (such as random tokens and delimiters), the LLM can be easily manipulated, effectively blinding or bypassing the automated SOC system.

## Chapter 3

### System Design and Implementation

This chapter presents the core architecture of SENTINEL. The two-tier design decisively resolves the performance bottlenecks associated with Generative AI in cybersecurity.

#### 3.1 Two-Tier Architecture Overview

SENTINEL segregates the processing pipeline into two specialized tiers (Figure 1). **Tier-1:** Operates at high-speed network edges, utilizing Regex rules and the Welford algorithm for anomaly detection. A Machine Learning Gateway (LightGBM) classifies and resolves the majority of alerts, escalating only complex, ambiguous cases to Tier-2. **Tier-2:** A locally hosted Agentic AI (Gemma-9B) references a knowledge base (RAG) and Threat Memory to make structured mitigation decisions (Block, Alert, Drop, HITL).

The operational flow (Figure 2) ensures transparency. Logs bypassing the Tier-1 filter are scored by the ML Gateway. Uncertain cases are securely enclosed within "Guardrails", enriched via RAG, and passed to the LLM for final adjudication.

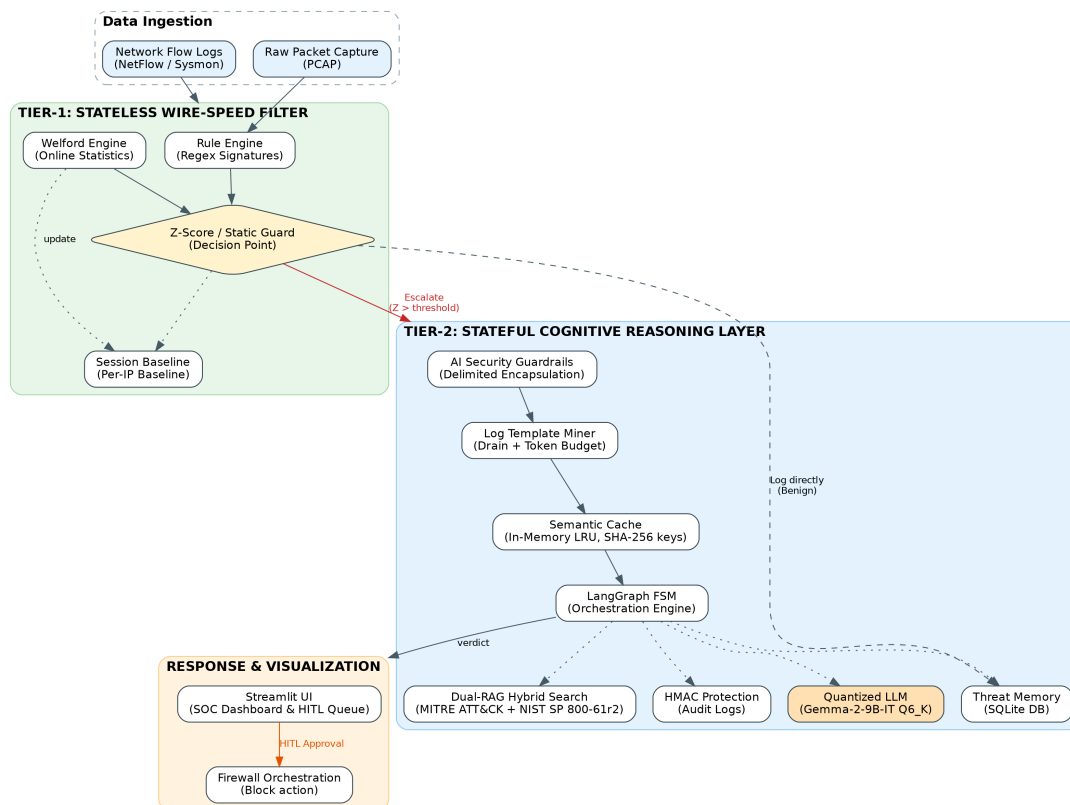


Figure 1. SENTINEL Overall Architecture: two-tier data flow and components. (Source: Author)

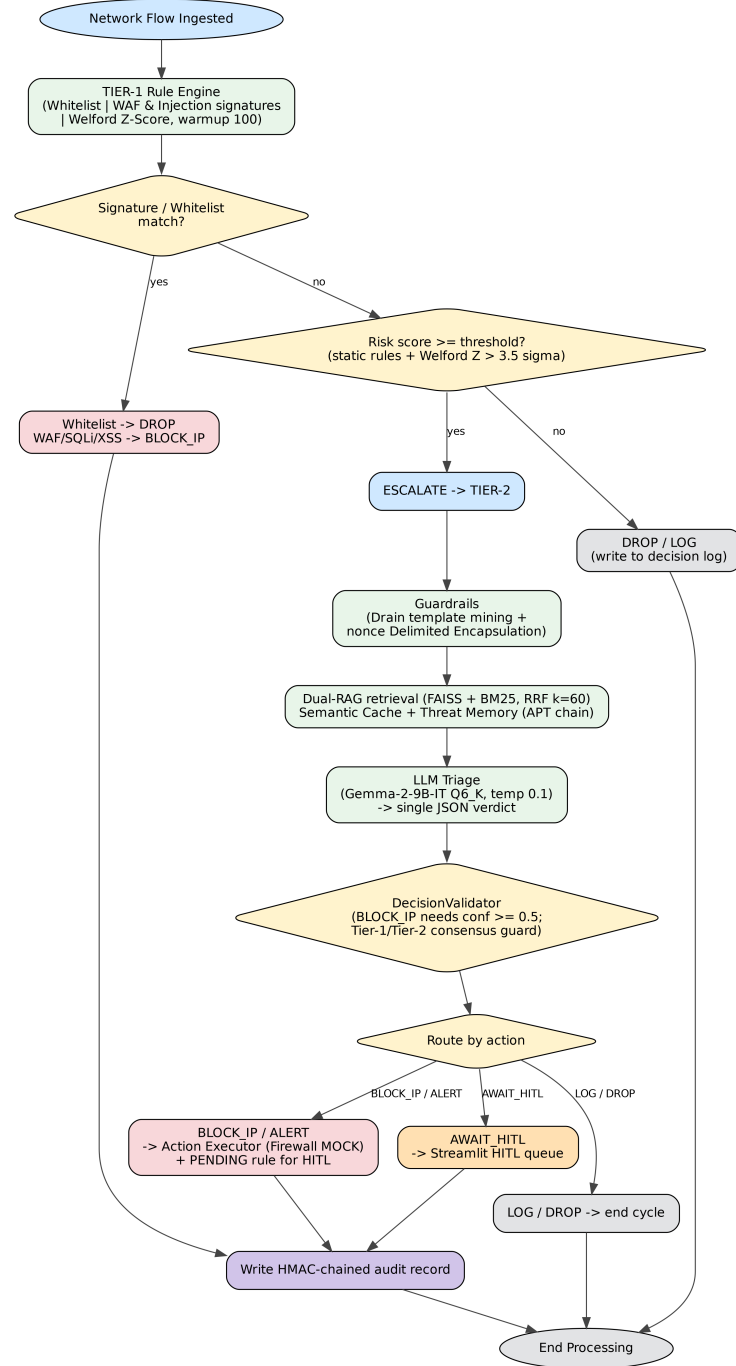


Figure 2. Detailed operational execution flow in the SENTINEL pipeline. (Source: Author)

### 3.2 Tier-1: Statistical Pre-filter & ML Gateway

Instead of forcing the LLM to process 100% of the traffic, Tier-1 absorbs over 80% of the volume.

- **(1) Welford Algorithm:** Computes real-time Z-scores per IP to detect anomalies without retaining entire datasets in memory.
- **ML Gateway (LightGBM):** Assigns risk scores. Extremely high-risk cases ( $> 0.85$ ) are blocked instantly, sparing LLM resources.
- **Semantic Cache (Tier 1.75):** Caches LLM resolutions for malicious payloads using SHA-256 hashes. Recurring payloads (e.g., Brute-force, DDoS) are blocked within 1 millisecond without LLM invocation.

### 3.3 Tier-2: Agentic AI and State Flow (LangGraph)

The LLM is orchestrated via a "State Graph" (LangGraph), functioning as a finite state machine (Table 1). This strictly limits AI hallucinations by enforcing rigid workflow milestones.

Table 1. State transition matrix of the Tier-2 cognitive agent. (Source: Author)

| State         | Input / Condition | Next          | Action              |
|---------------|-------------------|---------------|---------------------|
| GUARDRAILS    | Receive batch     | RAG_CONTEXT   | Compress & wrap     |
| RAG_CONTEXT   | Context retrieval | LLM_TRIAGE    | Add MITRE/NIST      |
| LLM_TRIAGE    | Action decision   | ATTACK_MAPPER | MITRE mapping       |
| ATTACK_MAPPER | BLOCK/ALERT       | ACTION_EXEC   | Block/Alert         |
| ATTACK_MAPPER | AWAIT_HITL        | HITL          | Escalate to Analyst |

### 3.4 Knowledge Retrieval (Dual-RAG) and Memory

To ensure highly specialized decisions, SENTINEL utilizes a hybrid RAG mechanism, fusing Vector search (semantic) and BM25 (keyword) via Reciprocal Rank Fusion (Table 2). The knowledge source comprises MITRE ATT&CK and NIST frameworks.



Table 2. Retrieval models in the hybrid search strategy (Dual-RAG). (Source: Author)

| Dimension | Dense (FAISS / Vector)          | Sparse (BM25 / Keyword) |
|-----------|---------------------------------|-------------------------|
| Model     | all-MiniLM-L6-v2                | BM25 Weights            |
| Strengths | Understands semantics, concepts | Exact IDs (CVE, IP)     |

**Threat Memory:** Used to correlate persistent attacks (APT) across time. IP reputation degrades over multiple days. If the score hits a critical threshold (100), the IP is automatically blocked at the network edge (Tier-1).

Table 3. Threat Memory Database Schema. (Source: Author)

| Table         | Primary Key                  | Purpose                    |
|---------------|------------------------------|----------------------------|
| ip_reputation | ip (PK),<br>reputation_score | Long-term IP reputation.   |
| threat_events | id, src_ip, label            | Multi-day APT correlation. |

### 3.5 Prompt Injection Defense and Transparency

Every payload ingested by the LLM is wrapped in random cryptographic nonces (e.g., «<DATA\_8F3E...>»). This signals the LLM that the enclosed content is purely passive data, forbidding the execution of any hidden instructions. Finally, every verdict is hashed (HMAC-SHA256) into an audit chain; any retrospective tampering breaks the chain, triggering immediate Dashboard alerts.

## Chapter 4

### Experiments and Evaluation

This chapter validates the SENTINEL architecture across five dimensions: Accuracy, Performance, Security, Explainability, and Integrity. The system is evaluated on consumer-grade hardware (Core i7, RTX 4060 Ti 16GB) using standard CSE-CIC-IDS2018 and DAPT2020 datasets.

#### 4.1 Tier-1 and ML Gateway Evaluation (Performance)

Tier-1 acts as a "steel shield" absorbing the vast majority of traffic. Table 4 demonstrates that Tier-1 achieves very high Precision (0.924) but intentionally moderate Recall (0.373), effectively deferring complex cases for deep analysis.

Table 4. Tier-1 Binary Classification (Aggregated Flow). (Source: Author)

| Metric           | Value |
|------------------|-------|
| Precision        | 0.924 |
| Recall (attacks) | 0.373 |
| F1-score         | 0.531 |

The offloading capability of the ML Gateway (LightGBM) is paramount (Table 5). It autonomously resolves 83.8% of suspicious logs in just 0.35 ms with 98.82% accuracy, rescuing the LLM from processing overload.

Table 5. LLM Offloading and ML Gateway Classification. (Source: Author)

| Metric                            | Value   |
|-----------------------------------|---------|
| Average Inference Latency         | 0.35 ms |
| LLM Offloading (autonomous)       | 83.8%   |
| Precision on autonomous decisions | 98.82%  |

## 4.2 Tier-2 Evaluation (Accuracy and Explainability)

For events that bypass the ML Gateway, the Tier-2 Agent processes them with absolute detection recall (Table 6). Not a single threat alert was overlooked (Conditional Recall = 1.00).

Table 6. Tier-2 Adjudication on escalated events. (Source: Author)

| Metric                                   | Value        |
|------------------------------------------|--------------|
| Conditional Threat Recall                | 1.00 (38/38) |
| Agent Confidence / JSON Resolution Error | 1.00 / 0     |

By accumulating events in the Threat Memory, SENTINEL can correlate APT campaigns (Table 7) with an average detection latency of only 8.33 events.

Table 7. Emergent multi-stage APT detection (DAPT2020). (Source: Author)

| Metric                    | Value       |
|---------------------------|-------------|
| APT IPs Detected / Missed | 3 / 0       |
| Average Detection Latency | 8.33 events |

Regarding Zero-day attacks lacking signatures, the Welford algorithm successfully escalated 12/15 attack classes by detecting flow anomalies (Table 8).

Table 8. Graded Zero-day detection frontier by standard deviation. (Source: Author)

| Deviation $k$ ( $\sigma$ ) | 2.0 | 3.0 | 3.5 (op.) | 4.0 |
|----------------------------|-----|-----|-----------|-----|
| Escalation Rate (%)        | 10  | 10  | 10        | 100 |

## 4.3 Ablation Study (Architectural Trade-offs)

The Ablation Study (Table 9) proves: Relying solely on the LLM (Config B) causes latency to spike to 14.9s PER event and triggers false alarms on 49% of clean traffic. The comprehensive two-tier architecture (Config F) maintains accuracy while keeping latency manageable (1.71s for complex cases).

Table 9. Controlled Balanced Ablation (150 benign + 150 attacks). (Source: Author)

| Configuration               | F1           | Prec.        | Rec.         | Escal.      | Lat./evt      |
|-----------------------------|--------------|--------------|--------------|-------------|---------------|
| A — Tier-1 Only (no LLM)    | 0.559        | 0.861        | 0.413        | —           | 0.04 s        |
| B — Pure LLM (no gates/RAG) | 0.804        | 0.673        | 1.000        | 100%        | 14.9 s        |
| <b>F — Full SENTINEL</b>    | <b>0.559</b> | <b>0.861</b> | <b>0.413</b> | <b>8.3%</b> | <b>1.71 s</b> |

## 4.4 Security and Lessons Learned

**1. Prompt Injection Resistance:** The static guardrail layer wrapping the Payload in nonce hashes enabled SENTINEL to block 100% of obfuscated attacks (Base64/Hex). Combined with the consensus guardrail, the system thwarted all attempts to hijack the LLM (Table 10).

Table 10. Static guardrail layer blocking rates on 120 payloads. (Source: Author)

| Attack Group                     | Block Rate   |
|----------------------------------|--------------|
| Encoding Bypass (Base64/Hex/URL) | 100.0%       |
| Structural / Delimiter Attacks   | 35.0%        |
| <b>Overall (Static)</b>          | <b>50.0%</b> |

**2. Fixing Evidence Stripping:** Initially, the system inadvertently stripped special characters from the Payload before feeding it to the LLM (mistaking them for delimiters). This blinded the LLM to SQLi/XSS attacks. **Solution:** Transmitting the raw Payload (including quotes and brackets) entirely and exclusively encapsulating it with normalized tags «<DATA\_BEGIN. . .>». SQLi detection rate instantly surged to nearly 99%.

**3. Fixing RAG Pollution:** When raw, noisy Payloads were fed directly into FAISS, retrieval results were skewed. **Solution:** Forcing the ML Gateway to affix a preliminary MITRE label (e.g., T1190) and combining it with BM25 Keyword Search, resulting in a dramatic increase in retrieval precision.

## 4.5 Integrity (LLM-as-a-Judge)

Cross-family evaluation by the Llama-3-8B model scoring the Gemma-2-9B agent averaged 3.1/5 (Table 11), with 100% of verdicts exhibiting no formatting hallucinations, indicating enterprise-grade stability.

Table 11. Cross-family LLM-as-a-Judge reasoning quality (Scale of 5). (Source: Author)

| <b>Dimension</b>                | <b>Average</b> |
|---------------------------------|----------------|
| Faithfulness (no fabrication)   | 3.30           |
| Answer Relevancy (action match) | 3.56           |
| <b>Total Average</b>            | <b>3.1 / 5</b> |

## 4.6 Summary

Experimental results prove that SENTINEL’s two-tier architecture offloads LLM processing by 83.8%, slashes latency by 82.97% compared to pure LLM usage, and ensures absolute immunity to log injection attacks.

## Chapter 5

### Conclusion and Future Work

#### 5.1 Conclusion

This thesis has proposed and validated **SENTINEL** - a two-tier cognitive architecture designed to solve the alert fatigue problem in Security Operations Centers (SOCs). By integrating a high-speed pre-filter (Tier-1) and a Machine Learning Gateway (LightGBM), the system successfully automated the resolution of 83.8% of network logs in merely 0.35 ms. This innovation slashed the end-to-end processing latency by 82.97% compared to standalone LLM deployments. For complex, ambiguous threats, Tier-2 leverages a locally hosted Agentic AI (Gemma-9B) combined with hybrid Retrieval-Augmented Generation (Dual-RAG) and Threat Memory to execute highly precise mitigations without violating Zero Trust security principles. Furthermore, the implementation of robust cryptographic Guardrails completely shielded the LLM from log-based Prompt Injection attacks, proving the enterprise-grade safety of the solution.

#### 5.2 Future Work

While the system has achieved promising results, several pathways remain to elevate SENTINEL to production-grade scalability. First, the architecture must transition to a Microservices model to enable horizontal scale-out on distributed systems like Kubernetes, replacing Redis Streams with Kafka to handle massive bandwidths without memory exhaustion. Second, the Tier-1 pre-filter could be upgraded with lightweight deep-learning models or re-implemented in system-level languages (Go/Rust) to push latency down to the microsecond threshold. Finally, a Multi-agent network could be developed to expand the system's capabilities beyond mere threat detection, fully automating digital forensics and deep incident response playbooks.

*Closing a humble endeavor, yet opening profound aspirations in the journey to secure the*

*digital realm through the power of Agentic AI.*

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